

Guns or Gandhi: Ethnic Elite Composition and Dissident Tactics in Civil Conflicts

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Abstract

This paper is very cool blbablablab

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Keywords: ethnic conflict, ethnic power relations, civil resistance, maximalist campaigns

Introduction

Theory & Hypotheses

Hypothesis 1. *A more fragmented ethnic ruling elite increases the likelihood that dissident campaigns adopt violent tactics over nonviolent tactics.*

Data and Methods

Data description

Our study employs a panel dataset of 480 so-called “maximalist campaigns”, a concept first introduced by Chenoweth and Stephan in *Why Civil Resistance Works* book (Chenoweth and Stephan 2011). Following their definition, a campaign is “a series of observable, continuous, purposive mass tactics or events in pursuit of a political objective”, with maximalist objectives encompassing regime change, self-determination, secession, and the expulsion of foreign occupations (Chenoweth and Shay 2019). We rely on Nonviolent and Violent Campaigns and Outcomes (NAVCO) 2.1 dataset, which records 384 maximalist campaigns from 1945 to 2013, extended by John Chin in his dissertation to 480 campaigns - the version we use (J. J. Chin 2017).

The central conceptual distinction is between violent and nonviolent repertoires of contention. Following NAVCO coding rules, nonviolent tactics ‘do not directly threaten or harm the physical wellbeing of the regime, its agents, or its citizens’ and the violent repertoire ‘involves the use of force to physically harm or threaten to harm the opponents’ (Chenoweth

and Shay 2019). The categories are mutually exclusive. A prominent example of a nonviolent campaign is the German 1989 *Die Wende* and the Cuban Revolution exemplifies a violent campaign. The violence/nonviolence distinction is well-established in the peace science literature and some significant findings have been made using NAVCO dataset (Celestino and Gleditsch 2013; Manekin and Mitts 2022; Cunningham 2013), some papers also rely on John Chin’s NAVCO 2.1 extension, which we use (J. Chin, Song, and Wright 2023). Violence is operationally coded as a binary variable (1 = a campaign takes a violent form, 0 = a campaign takes a nonviolent form).

Our key independent variables are drawn from the Ethnic Power Relations (EPR) Core Dataset (Vogt et al. 2015). EPR provides information on ethnic groups and their relative access to power. Following Hendrix and Salehyan (2019), we operationalize the ethnic ruling elite fragmentation via the *Included fractionalization index (IFI)* and the *Excluded fractionalization index (EFI)*. IFI is “a Herfindahl-like index of all groups counted as ‘included’ (i.e. having access to executive power) during a given country-year”, whilst EFI is “a Herfindahl-like index of all groups counted as ‘excluded’ during a given country-year”. Thus, a higher IFI indicates a more concentrated ethnic ruling elite and a lower IFI shows that ethnic power is distributed more equally, across many groups. Instead of relying on raw counts, IFI and EFI are more theoretically motivated. If a raw count treats two countries with the ethnic groups in power count of, say, 3 equally, IFI and EFI account for ethnic power distributions.

To address endogeneity and block non-causal paths, we control for a standard set of confounders: GDP per capita in constant PPP dollars (Gapminder 2024); GDP annual growth rates; logged population size (Nations 2024); youth bulge - the proportion of young

population (15-29) to the total population size; executive corruption scores (Coppedge et al. 2024); polyarchy (electoral democracy) scores (Coppedge et al. 2024); regime duration in years (Coppedge et al. 2024); peace years (years elapsed since the last maximalist campaign in each country); EFI to account for a possible confounding of excluded ethnic groups as well. To address reversed causality, all the independent variables are one-year lagged¹.

Descriptive statistics

Table 1: Ethnic and political attributes of NAVCO 2.1 campaign onsets

	Nonviolent (N=245)		Violent (N=235)	
	Mean	Std. Dev.	Mean	Std. Dev.
Included Fractionalization Index	0.405	0.308	0.300	0.285
Excluded Fractionalization Index	0.043	0.093	0.072	0.152
ln(GDP per capita)	8.717	0.898	7.989	0.949
ln(Population)	9.810	1.603	9.890	1.567
Youth Bulge	42.107	7.960	45.593	6.194
Polyarchy	0.338	0.197	0.278	0.173
Executive Corruption	0.616	0.244	0.627	0.242
ln(Regime Duration)	2.229	1.308	1.685	1.243
Peace Years	10.691	13.771	7.826	10.296

Note: All variables one-year lagged.

Table 1 presents descriptive statistics for all variables included in the analysis between nonviolent and violent campaigns. The table shows a balance between violence and nonviolence instances. Nonviolent campaigns have a slightly higher mean Included Fractionalization Index value, while violent campaigns face a slightly higher mean Excluded Fractionalization Index value. For control variables, such as GDP and polyarchy, the table shows some degree of variation between violent and nonviolent campaigns. In general, data missingness is not a

¹Even though we find it unconvincing that a choice of one tactic over the other could affect IFI and EFI

significant issue, though there is missing data on ethnic groups for all campaigns in countries where data collection is complicated and ethnic lines are profound, such as Yemen and Somalia². Figure 1 shows the distribution of the included fractionalization index with respect to the tactic choice, visualized by both a histogram and a density plot. At low IFI values, violent tactics are more frequent and with the growing IFI the nonviolent tactic is prevalent. However, with the growing IFI both tactic choices occur less frequently, reflecting the mass anti-regime mobilization logic - with the growing elite cohesion, political opportunities of rebellion are decreasing, regardless of the repertoire of contention. This visualization shows some degree of variation of 1) the Included Fractionalization Index 2) between violent and nonviolent choice.

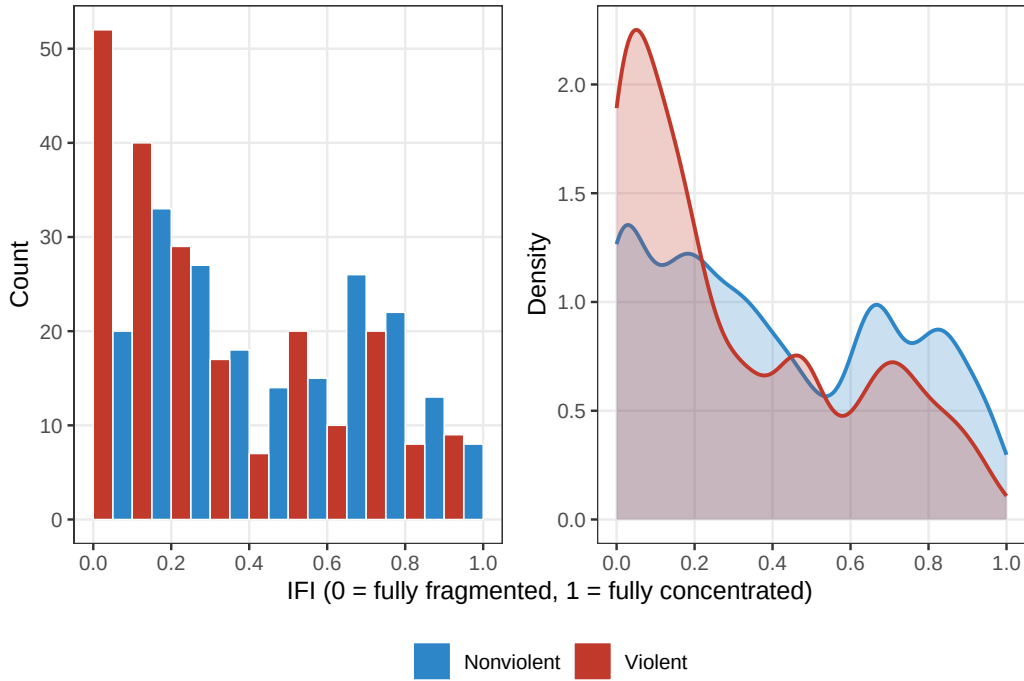


Figure 1: Included fractionalization index by Repertoire Choice: a histogram and a density plot

²In general, only 6.46% of IFI values and 6.46% of EFI values are missing

Modeling approach

Given a binary dependent variable and a balanced sample -235 of violent and 245 of nonviolent onsets - we estimate a logit model. As we do not predict the duration of events, there is no need for advanced binary choice models such as a complementary log-log. To account for a nested structure of our data, where campaigns are nested within countries, we add a country-level random varying intercept in all models.

Including a country fixed effect to control for the time-invariant unobserved factors such as culture or geography is theoretically appealing and common in the peace science literature (Butcher and Svensson 2016; Ives and Breslawski 2022; Fearon and Laitin 2003; Jung 2025). However, given that ethnic structures are almost constant *within* countries and vary predominantly *between* countries - the variation that a country fixed effect absorbs, adding it is methodologically inappropriate. Exploratory data analysis showed that the median within-country standard deviation of both IFI and EFI rounds to 0 (up to two decimal points without rounding), meaning that the FE identification would test on a very small sample of countries that experienced ethnic configuration changes between maximalist campaigns. Random effects exploit both within- and between-group variations, which is appropriate for our research question and data structure. Moreover, a random-effects estimator introduces “a compromise between the fixed-effects and pooled models”, though we acknowledge that the coefficients can be partially biased in case varying intercepts correlate with the regressors (Clark and Linzer 2015). Despite this potential bias, RE is still a more preferable choice over FE.

For Hypothesis 1, our model specification is:

$$\text{violent}_{it} \sim \text{Bernoulli}(\pi_{it})$$

$$\log\left(\frac{\pi_{it}}{1 - \pi_{it}}\right) = \beta_0 + \beta_1(\text{IFI})_{it-1} + \beta_2(\text{EFI})_{it-1} + \mathbf{X}'_{it-1}\boldsymbol{\gamma} + \alpha_{[j]i}$$

$$\alpha_j \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$$

where $\mathbf{X}_{it-1}$ is the vector of covariates and $\alpha_{[j]i}$ is a country varying intercept.

For uncertainty calculations, we follow a King’s simulation approach (King, Tomz, and Wittenberg 2000). This approach is also well-established in peace science literature (Butcher and Svensson 2016). Instead of reporting coefficients and their significance, we calculate predicted probabilities as the main quantity of interest via an observed value approach (OVA), meaning we simulate the predicted probabilities for all observations and then calculate the mean of them³. We simulate over the full available range of the focal variable values. For the essay, we define `sim_ova_range` function for the OVA simulations, allowing to calculate predicted probabilities in just four lines of code. It also allows for different CI levels, seeds, number of values and focal variable ranges at the researcher’s discretion.

Results

To test the first hypothesis, we estimate a logit model of the violent tactic choice, according to the H1 model specification. Figure 2 shows the simulated predicted probabilities of a violent tactic choice over a full available range of observed IFI values. Despite a downward trend - from 0.46 at IFI equal to 0 to 0.42 at IFI equal to 0.96 - confidence intervals that

³We report coefficients only in the cross-validation robustness check

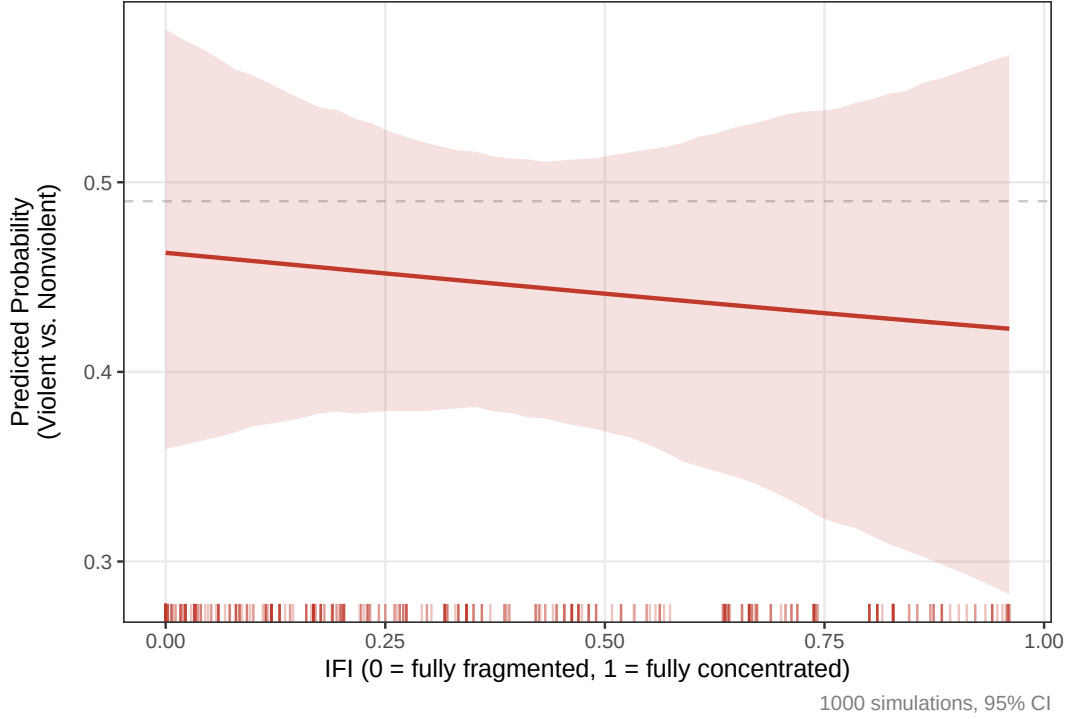


Figure 2: Simulated predicted probabilities of a violent tactic choice over a range of included fractionalization index values

encompass 0.5 do not allow us to claim that with the increasing IFI the rebels would prefer violence over nonviolence. Instead of zero probability, our benchmark probability is approx. 0.49 - the sample proportion of violent campaigns in the whole sample.

Table 2: Confusion Matrix and Predictive Performance — Model 1

	Actual Nonviolent	Actual Violent
Confusion Matrix		
Predicted Nonviolent	45	20
Predicted Violent	15	44
Performance Metrics		
Accuracy	0.718	
Precision	0.746	
Recall	0.688	
F1 Score	0.715	
AUC	0.777	

Despite an insignificant effect of IFI on a violent choice, the model as a whole demon-

strates a strong predictive performance. We divided the dataset into a train and test sample, estimated a logit model on the train sample and predicted a tactic choice on the test set. Table 2 reports the confusion matrix and performance metrics. With an accuracy of 0.72 and F1 score of 0.72, our model in general predicts tactic choice well above a coin toss, predicting violence a bit better than the nonviolent choice. Another measure of model fit is the area under the ROC curve (AUC), showing whether our model predictions of a violent choice are better than a random draw. The AUC score of 0.78 indicates a good model fit. Figure A1 plots an AUC-ROC curve for this model. An important implication here is that the model as a whole (including the specification itself, a set of covariates and a country varying intercept) captures meaningful variation in the outcome, even though the IFI does not significantly predict a violent choice. Such results reflect a real absence of the IFI effect rather than a model misspecification.

Robustness checks

We conduct four robustness checks. First, given that civil conflicts and a tactic choice are regionally clustered, we estimate an additional model on the Middle East and North Africa (MENA) and Sub-Saharan Africa countries subsample, where violent campaigns are more expected than in other regions. For the full sample, there are 49% of violent campaigns, whilst for MENA & Sub-Saharan Africa there are 58.3% of them. Second, territorial campaigns could be more ethnicity-driven and secessionist civil wars along ethnic lines are widely researched in the academic literature (Collier and Hoeffler 2006; Denny and Walter 2014; Toft 2012). For this reason, we estimated three separate models 1) on the MENA & Sub-Saharan Africa subsample, 2) excluding territorial campaigns and 3) both on the MENA & Sub-

Saharan Africa sub-sample with excluded territorial campaigns. Finally, following Bell and Jones (2015), we estimate an alternative Within-Between Random Effect (REWB) model specification. REWB presents a main independent variable in two forms: its demeaned form and a group mean. De-meaning allows to estimate a *within* (essentially fixed) effect and save the underlying assumption of the varying intercept model. An important note here is that we do not compare the model fit of these specifications because of the different samples, which make the models incomparable in terms of model fit measures.

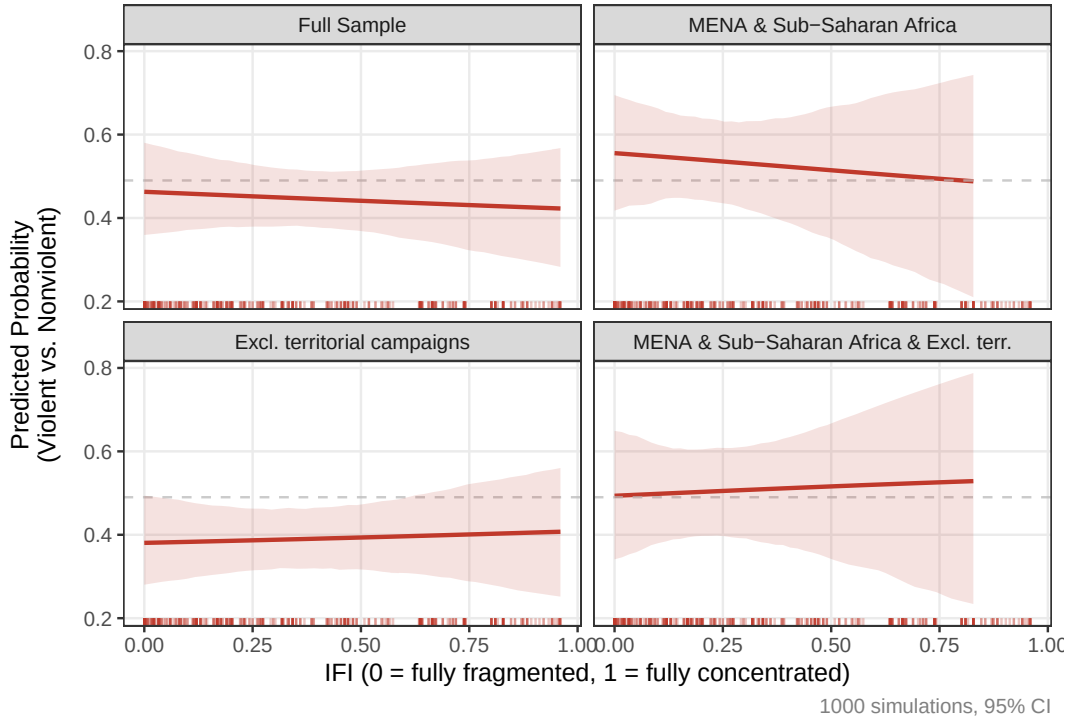


Figure 3: Simulated predicted probabilities of a violent tactic choice over a range of IFI values - robustness checks results

We calculate simulated predicted probabilities over the full available range of IFI values for all three robustness check models and summarize the findings on one plot⁴. Figure 3

⁴An important note here is that a model with excluded territorial campaigns did not initially converge. To address this, we reestimated the model with all the available optimizers with a warm start, with the `bobyqa` optimizer demonstrating a successful convergence. We use this reestimation for simulating predicted probabilities.

shows the results along three robustness specifications, including the full-sample baseline. The only partially significant finding is in the model which excludes territorial campaigns (this sub-sample mostly includes anti-regime campaigns). For low IFI values, campaigns that exclude territorial goals/claims have a higher chance of being nonviolent and the rebels slowly push towards a higher likelihood of violence with the growing IFI value. This (though cautious) finding can be explained by the logic of mass anti-regime mobilization, where a more concentrated ethnic elite decreases the benefits of nonviolent resistance and pushes the rebels toward violent repertoires of contention. For the MENA & Sub-Saharan Africa subsample (both with and without territorial campaigns) there is no systematic difference between violence and nonviolence across the growing IFI value. Figure A2 plots simulated predicted probabilities for the baseline and the REWB model, demonstrating practically identical results (see Appendix A). In general, our null result is robust⁵.

Table 3: Model 1: Main vs. Cross-Validated Coefficients

Variable	Main Model	Cross-Validated
IFI	-0.206 (0.671)	-0.385 (0.438)
EFI	1.781 (1.482)	1.859 (1.696)
ln(GDP per capita)	-1.066 (0.260)	-0.765 (0.176)
ln(Population)	-0.027 (0.127)	0.058 (0.101)
Youth Bulge	0.015 (0.031)	0.019 (0.018)
GDP Growth	-0.021 (0.015)	-0.025 (0.005)
Executive Corruption	-0.184 (0.690)	-0.392 (0.230)
Polyarchy	0.212 (0.929)	-0.250 (0.723)
ln(Regime Duration)	-0.361 (0.123)	-0.322 (0.044)
Peace Years	0.009 (0.012)	0.001 (0.003)

Note:

Intercept is not reported. Standard errors in parentheses.

Cross-validated SE via King (2001) combination rule.

⁵We do not calculate performance metrics for the robustness check specifications due to identification issues and low sample sizes, caused by simultaneous regional/campaign type and train-test subsampling.

Finally, we cross-validate the results (5 cross-validations) to check whether the same results hold when models are estimated on different random subsamples. For this, we define the `cross_validation_fun` function⁶. For this part, we did not calculate quantities of interest, but rather compare coefficient values and their significance, which will be enough to judge whether the results are stable or not. Table 3 shows the coefficients differences between the main and the cross-validated model. For both models, the IFI coefficient is negative and insignificant, though the coefficient absolute values differ. Moreover, one could notice a slight inflation of the standard errors, though not significantly changing the null result. Moreover, the reader should account for the sample size and respective test sample sizes that partly explain standard errors' inflation.

Discussion

Conclusion

⁶Cross-validation subsampling causes convergence issues due to low sample sizes of the subsamples. To address this, we force the function to resample and reestimate the model until convergence. Thus, it can take more than one attempt to estimate one cross-validated subsample. It took 36 attempts to estimate 5 cross-validations

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Appendix

Appendix A: AUC - ROC Curve

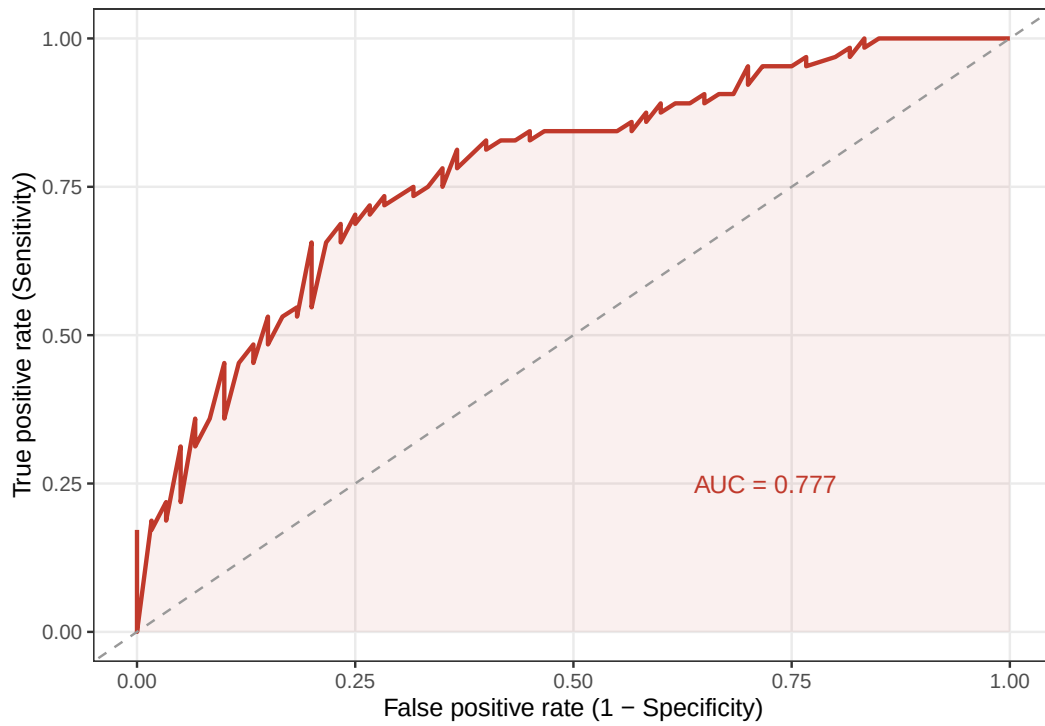


Figure A1: AUC-ROC Curve - Model 1

Appendix B: Within-Between Random Effects Robustness Check

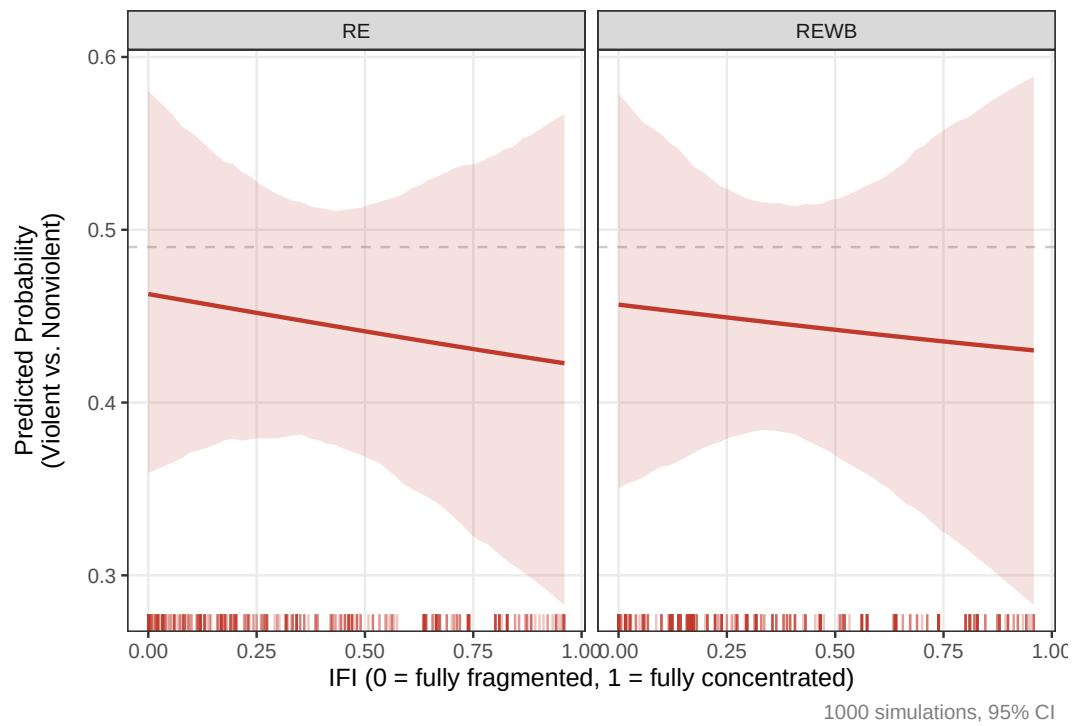


Figure A2: Simulated predicted probabilities of a violent tactic choice over a range of IFI values - baseline vs. REWB